

Sebastian Salinas

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EDUCATION

Texas Tech University, Lubbock, Texas GPA: 4.0

Bachelor of Science in Biochemistry May 2025

Summa Cum Laude | Minor: Biology

Coursework: Genetics, Biochemistry I-III, Biophysical Chemistry, Physical Chemistry I-II, Immunology, Organic Chemistry I-II, Microbiology, English 1 & 2

- Tech10 (2025)

Presidents List (2021-2025)

Extra Kilometer Award (2024)

Outstanding Student Leadership Award (2025)

CURRENT RESEARCH

J20 Alzheimer's Mouse Model Lab | PI: Dr. Josh Lawrence, TTUHSC May 2025 –

Present

Research Technician

Investigate the effect of epigenetic and metabolic interventions on neurodegeneration in the J20 transgenic Alzheimer's mouse model. All-trans retinoic acid (ATRA), a vitamin A derivative, and the HDAC inhibitor vorinostat have demonstrated neuroprotective effects. These compounds halt and/or reverse neurological deterioration in mammalian neuronal tissue through mechanisms that remain under active investigation. Research projects focus on dietary vitamin A supplementation in the form of ATRA and pharmacological HDAC inhibition with vorinostat and determining their effects on neurophysiology, excitability, electrophysiology, behavior, morphology, and biochemistry. Current work integrates qRT-PCR, Western blot, histology, microscopy, behavior, patch clamp, and dietary and pharmacological interventions to investigate how ATRA and vorinostat modulate Alzheimer's-related phenotypes. Supported by a Culture of Care Award from the Garrison Institute on Aging. Presented findings at SFN 2025.

RESEARCH TRAINING

Keyel Lab, Texas Tech University, Lubbock, Texas November 2022 – May 2025

Undergraduate Researcher | Advisor: Dr. Peter Keyel

- Examined how cholesterol and sphingolipid-rich plasma membrane microdomains in *Leishmania major* influence susceptibility to pore-forming toxins.
- Achieved LC50 values for streptolysin O (SLO), a cholesterol-dependent cytolysin (CDC), as well as preformed binding assays on non-CDC variants of the toxins.

- Assessed surface coat proteins as modulators of cytolytic susceptibility, with lipophosphoglycan and 14-methylated sterols protecting against SLO-mediated pore formation.
- These findings advance understanding of membrane repair and have implications for host-pathogen interactions, wound repair, and membrane biophysics. Presented findings at 2 ASCB meetings.

Cardiovascular Mechanics Lab, Lubbock, Texas *December 2021 – May 2022*

Volunteer Researcher

- Assisted in designing cardiovascular heart valves under Dr. Zhaoming He.
- Conducted advanced literary analysis and meta-analyses using online research databases and Blood flow analysis via MATLAB.

NASA Space and Life Sciences Institute, Houston, Texas *May 2022 – August 2022*

Research Intern

- Observed studies on the effects of lunar dust and simulants on lung epithelial cells in Dr. Jeeverajan's lab, employing fluorescence microscopy and chemical modeling software.
- Developed strategies to mitigate contact with lunar materials.

Johnson Space Center, NASA, Houston, Texas *April 2018 – May 2020, May 2023 – August 2023*

Technical Intern

- 18th Annual Space Life Sciences Summer Institute intern. Technical internship and project design in Energy Systems design.
- Use of computer design software and CAD programs (such as Inventor).

Tutor | Dept. of Biological Sciences, Texas Tech University, Lubbock, Texas *January 2022 – May 2022*

- Tutor for introductory biology lab, covering molecular biology, genetics, classical and molecular evolutionary biology, and ecology and environmental biology.

TEACHING

Teaching Assistant, Confocal Microscopy – GSBS Masters Program, Dept. of Pharmacology and Neuroscience, TTUHSC *Sept 2026 – Present*

Graduate Teaching Assistant to Dr. Josh Lawrence

- Assist Dr. Josh Lawrence with masters-level confocal microscopy instruction for GSBS students.
- Guide a small cohort (4 in first session, 2 additional expected) through hands-on scope time: multichannel imaging, acquisition, and ZEN software.
- Support sample preparation and quantitative image analysis; deliver introductory lecture on confocal principles.

PUBLICATIONS

1. CS Haram, **SJ Salinas**, C Wilson, SW Sheikh, K Zhang, & PA Keyel. (2026) (Publication), *Functional Dissection of Leishmania major Membrane Components in Resistance to Cholesterol-Dependent Cytolysins*. Toxins. 2026; 18(1):46.
<https://doi.org/10.3390/toxins18010046>
2. **S. Salinas**, C. Haram, K. Zhang, & P. Keyel (2024) (Poster). *14-methylated sterols protect the human pathogen Leishmania major against membrane damage by cholesterol-dependent cytolysins*. Mol. Biol Cell 36. Abstract #2057. American Society of Cell Biologist (ASCB) Cell Bio 2024
3. C. Haram, R. Keane, **S. Salinas**, S. Waheed, K. Zhang, P. Keyel (2024). *Leishmania major is an unconventional system to study membrane damage by bacterial toxins*. Mol Biol Cell 36. Abstract #2944
4. **S. Salinas**, C. Haram, K. Zhang, & P. Keyel (2023) (Poster). *Non calcium ions regulate membrane repair in human pathogen Leishmania major*. Mol. Biol Cell 35. Abstract #2900. ASCB Cell Bio 2023
5. C. Haram, R. Keane, **S. Salinas**, K. Zhang, P. Keyel (2023). *Lipophosphoglycan shields Leishmanial GPI anchored proteins from the pore forming toxin aerolysin*. Mol Biol Cell 35. Abstract #1561
6. **S. Salinas**, C. Haram, S. Waheed, K. Zhang, P. Keyel (2024) (Talk). *Leishmania is a system to study cell survival*. TrUE Symposium, Abstract #4d1. Texas Tech Center for Transformative Undergraduate Experiences.
7. **S. Salinas**, C. Haram, K. Zhang, P. Keyel (2023) (Poster). *Detailing mechanism behind Membrane Repair in Leishmaniasis*. TTU Undergraduate Research Conference, Abstract #A174. Discoveries to Impact 2023.

IN PREPARATION

1. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Retinoic acid and HDAC Inhibition reprogram oxidative stress responses via context-dependent antioxidant and transcriptional mechanisms in cultured neurons*. Working draft.
2. Baker, A., **Salinas, S.**, Lie, P., Smith, S.C., Skawratananond, S., Bose, C., & Lawrence, J.J. *KCNIP1 at the Convergence of Nutritional Deficiency, Ion Channel Dysfunction, and Alzheimer's Disease*. Working draft.

ACCEPTED ABSTRACTS

1. Lie, P.E., Baker, A., Self, D., Khanna, N., Zhang, C., **Salinas, S.**, Smith, S.C., Skawratananond, S., Kisby, B.R., Shanmugam, S., Ponomarev, I., Telias, M., & Lawrence, J.J. *Dopaminergic Hippocampal Interneuron Associated with Cognitive Severity Links Retinoic Acid Signaling, and Synaptic Scaffolding Failure in the J20 Alzheimer's Mouse Model*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.

2. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of Dietary Vitamin A in Dentate Gyrus Granule Cell Excitability in a Mouse Model of Alzheimer's Disease*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.
3. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A-Dependent Modulation of Behavioral and Biochemical Profiles in an Alzheimer's Disease Mouse Model*. Alzheimer's Association International Conference, London, United Kingdom, July 12th-15th, 2026.
4. **Salinas, S.**, Baker, A., Smith, S.C., Strickland, J., Bose, C., & Lawrence, J.J. *Long-Term Vitamin A Supplementation Decelerates Cognitive Deficits and Anxiety-Like Phenotypes in the J20 Alzheimer's Mouse Model*. Society for Neuroscience Conference, Washington, DC, November 14th-18th, 2026.
5. Baker, A., Smith, S.C., **Salinas, S.**, Bose, C., Skawratananond, S., Tanner, N., & Lawrence, J.J. *Behavioral and Cellular Effects of Vitamin A Deficiency in a Mouse Model of Alzheimer's Disease*. Alzheimer's Disease International Conference, London, United Kingdom, July 12th-15th, 2026.

POSTER PRESENTATIONS

1. Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A supplementation modulates behavioral and biochemical profiles in a mouse model of Alzheimer's disease*. CTNT/GIA, TTUHSC, April 2026
2. Lie, P.E., Baker, A., Khanna, N., Self, D., Chakroborty, A., Smith, S.C., **Salinas, S.**, Skawratananond, S., Kisby, B.R., Shanmugan, M., Telias, M., Ponomarev, I., & Lawrence, J.J. *Computational ethology characterizes behavioral inflexibility and excitatory/inhibitory transcriptional dysregulation in vitamin A deficient Alzheimer's disease model mice*. CTNT/GIA, TTUHSC, April 2026
3. Luong, K., Baker, A., Lie, P., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A Supplementation Reduces Anxiety and Improves Spatial Cognition in the J20 Mouse Model of Alzheimer's Disease*. CTNT/GIA, TTUHSC, April 2026
4. Kim, A., Zhang, C., **Salinas, S.**, Hindle, A., Baker, A., Smith, S.C., Singh, S.P., Lawrence, J.J., & Bose, C. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in hippocampal neurons*. Undergraduate Research Conference, Texas Tech University, April 2026
5. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A on dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. 9th Annual Symposium on Brain Health, GIA, TTUHSC, April 2026
6. Baker, A., **Salinas, S.**, Henderson, M., Smith, S.C., Skawratananond, S., Bose, C., Logsdon, A., & Lawrence, J.J. *Vitamin A Supplementation Reduces Anxiety and Improves Spatial*

Cognition in the J20 Mouse Model of Alzheimer's Disease. Texas Alzheimer's Research and Care Consortium, Austin, TX, January 2026

7. Baker, A., Smith, S.C., **Salinas, S.**, Bose, C., Skawratananond, S., Tanner, N., & Lawrence, J.J. *Behavioral and cellular effects of vitamin A deficiency in a mouse model of Alzheimer's disease*. Alzheimer's Disease International Conference, London, United Kingdom, July 12th-15th, 2026

8. Lie, P.E., Baker, A., Khanna, N., Self, D., Chakroborty, A., Smith, S.C., **Salinas, S.**, Skawratananond, S., Kisby, B.R., Shanmugan, A., Hindle, M., Telias, M., Ponomarev, I., & Lawrence, J.J. *Machine learning demonstrates vitamin A-mediated partial rescue of behavioral phenotype in an Alzheimer's disease model mouse*. Student Research Week, TTUHSC, February 2026

9. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Sex differences in dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. Student Research Week, TTUHSC, February 2026

10. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A on dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. First Annual Meeting on Human Molecular Aging, Innovation Hub at Research Park, April 2026

11. Self, D., **Salinas, S.**, Zhang, C., & Lawrence, J.J. *Roles of dietary vitamin A in dentate gyrus granule cell excitability in a mouse model of Alzheimer's disease*. 38th Annual Student Research Week, TTUHSC, February 2026

12. Kim, A., Zhang, C., **Salinas, S.**, Hindle, A., Baker, A., Smith, S.C., Singh, S.P., Lawrence, J.J., & Bose, C. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in hippocampal neurons*. 38th Annual Student Research Week, TTUHSC, February 2026

13. Kim, A., Zhang, C., **Salinas, S.**, Baker, A., Smith, S.C., Singh, S.P., Bose, C., & Lawrence, J.J. *All-trans retinoic acid and vorinostat synergize to modulate histone acetylation and oxidative stress in the hippocampus of a mouse model of Alzheimer's disease*. Student Research Week, Texas Tech University, April 2026

SKILLS

Laboratory Techniques:

- 4 years of experience: PCR, Immunohistochemistry, Confocal Microscopy, Widefield fluorescence microscopy.
- 3 years of experience: Flow Cytometry, Western Blotting, Leishmania Promastigote Cell Culture, SDS-PAGE.
- 2 years of experience: High Performance Liquid Chromatography, Column chromatography, Mass spectroscopy.
- 1 year of experience: Mammalian Cell culture, Transformation, Plasmid Isolation, Protein Purification, Mouse colony maintenance, Rodent handling, Rodent Tissue collection.

Imaging & Analysis:

- Volume rendering, segmentation, tracing, digital reconstruction, 3D visualization, quantitative analysis of neuronal and subcellular structures.

Software & Tools:

- ImageJ, Imaris, fluorescence quantification pipelines, MATLAB, FlowJo, Benchling, GraphPad Prism, Adobe Photoshop, Biorender. Blood flow analysis via MATLAB.

Techniques:

- DNA/RNA extraction, qRT-PCR, spectrophotometry, gel electrophoresis, ELISA, protein biochemistry, sterile technique, tissue processing, cryosectioning, histology, immunofluorescence staining, cell culture, CRISPR-Cas9, Solution and buffer preparation, chemical handling, sample preparation, lab equipment operation and maintenance, mechanical and chemical tissue disaggregation.

Research:

- Experimental design, cohort selection, blinding, randomization, data analysis, data management, scientific writing, lab meetings, literature reviews, hypothesis formulation, record-keeping, manuscript writing.

Certifications:

- CITI human subjects, CITI animal use, OSHA bloodborne pathogens.

HONORS AND LEADERSHIP

Tech10 recipient, Texas Tech University *May 2025*

President, Raider Recycling *Aug 2023 - May 2025*

- Reinitiated recycling collection in residence halls across the Texas Tech Campus.
- Lead a team of 8 officers to enhance sustainability projects and organize recycling initiatives at Texas Tech.

Outstanding Student Leadership Award, Center for Integrated STEM and Education Research *April 2025*

Mentor, Honors College Mentor Program, Texas Tech University *May 2020 - March 2025*

Extra Kilometer Award, Center for Integrated STEM and Education Research *April 2024*

Texas Tech Association of Biologists, Active Member *Aug 2024 - May 2025*

- Participate in organizational volunteering activities as well as social activities.

CISER (Center for Integration of STEM Education & Research), Active Member *Mar 2021 - May 2025*

- Conduct science outreach for middle and elementary school students.

REFERENCES

Dr. J. Josh Lawrence, Ph.D. Dept. of Pharmacology and Neuroscience, Texas Tech University Health Sciences Center, John.Lawrence@ttuhsc.edu, 806-743-2656

Dr. Chhanda Bose, Ph.D. Dept. of Pharmacology and Neuroscience, Texas Tech University Health Sciences Center, chhanda.bose@ttuhsc.edu, 806-743-2801

Dr. Peter Keyel, Ph.D. Dept. of Biological Sciences, Texas Tech University, peter.keyel@ttu.edu, 806-834-8241

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